Module 3: Mathematical Physics II

Sequential numbers

A sequence is a collection of elements with a particular order.

Elements of a sequence can be related to each other, and are often defined using recursion.

Sequences can be finite or infinite.

$$f_n = f_{n-1} + f_{n-2}$$

The Fibonacci Sequence

1,1,2,3,5,8,13,21,34,55,89,144,233,377…

1+1=2
1+2=3
2+3=5
3+5=8
5+8=13
8+13=21
13+21=34
21+34=55
34+55=89
55+89=144
89+144=233
144+233=377

Sequence can be defined as a function.

Random numbers

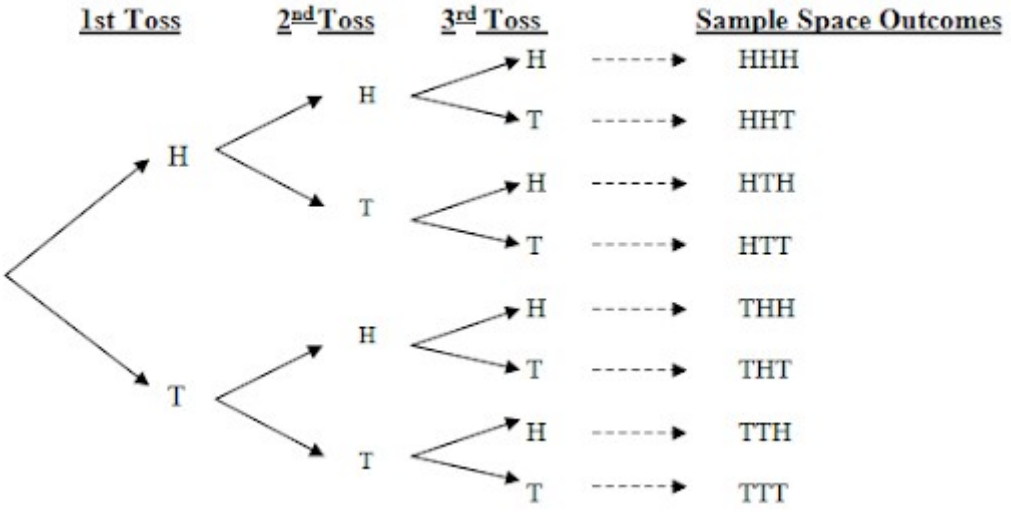
A random number is a number chosen as if by chance from some specified distribution.

Such numbers are independent, having no correlations between successive numbers.

Probability

Probability is the possibility of occurrence of an event or outcome.





Sample Space: In probability, the set of all possible outcomes is called the Sample Space. We will use S to represent the sample space. In terms of the language of sets, a sample space is a universal set and an outcome is an element of the universal set.

Probability

The classical probability P(A) for an event A is defined by the number of favorable results n, divided by the number of possible results m

$$P(A) = \frac{n}{m}$$

$$P(\bar{A}) = 1 - \frac{n}{m}$$ Probability of complementary event

The statistical definition of the probability for an event A is given by:

$$P(A) = \lim_{m \to \infty} \frac{n}{m}$$

Random numbers

Random numbers generated in the computer are pseudo-random numbers.

A pseudo-random number is a number generated with the help of a deterministic algorithm, however, it shows a behavior as if it were random.

A random number generator will have to comply with following criteria

1. Produce pseudo-random numbers whose statistical properties are as close as possible to those of real random numbers.

2. Have a long period: It should generate a non-repeating sequence of random numbers which is sufficiently long for computational purposes.

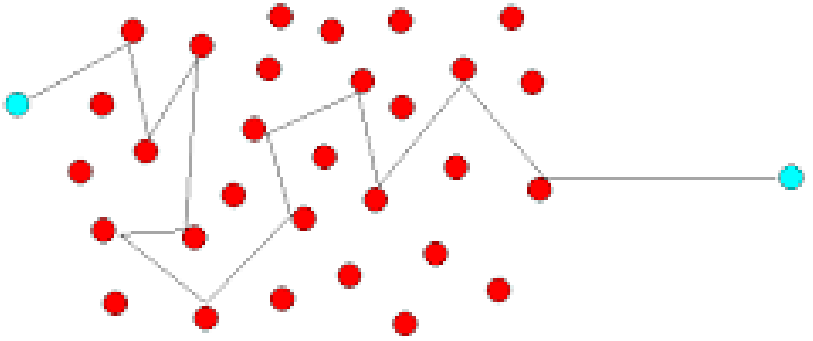
3. Be reproducible in the sense defined above, as well as re-startable from an arbitrary break-point.

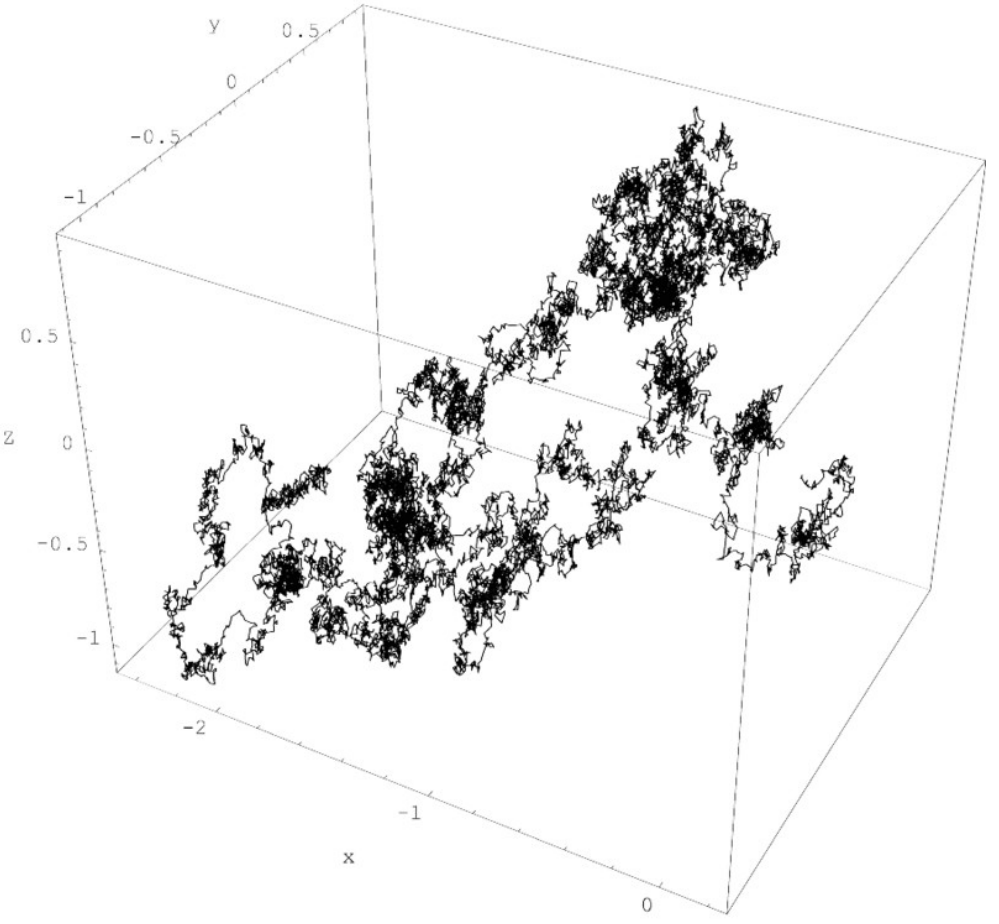
4. Be fast and parallelizable: It should not be the limiting component in simulations.

Randomness

Randomness: It is the lack of predictability of events, i.e., there is no pattern involved.

A random process is non-deterministic.





Random variable

A random variable is a variable whose value is unknown or a function that assigns values to each of an experiment's outcomes.

A random variable can be either discrete (having specific values) or continuous (any value in a continuous range).

The use of random variables is most common in probability and statistics, where they are used to quantify outcomes of random occurrences.

Risk analysts use random variables to estimate the probability of an adverse event occurring.

Random variable

A random variable is a functional whose value is assigned to the outcome of some random process.

For an event ω, random variable can be defined as X(ω)

$$\langle X \rangle = \sum X(\omega)\, p(\omega)$$

Expectation value of X

Probability of event ω

$$var(X) = \langle X^2 \rangle - \langle X \rangle^2$$

$$\sigma = \sqrt{\langle X^2 \rangle - \langle X \rangle^2}$$

Standard deviation

Random variable

In a random process, to ascertain the outcome of an event or to determine the <X>, multiple set of observations are utilized

$$x = \langle X \rangle$$

For a set of observations x_i where i = 1; 2; ..... ; N

$$\bar{x} = \frac{1}{N} \sum_{i=1}^{N} x_i$$

Central limit theorem

Let X_1, X_2, X_3, ..., X_n are the n random samples with mean μ and variance σ^2

Then the limiting from of the distribution

$$Z_n = \frac{\sum_{i=1}^{n} X_i - \mu}{\sqrt{n\,\sigma^2}}$$

tends to be normal.

Central tendencies and dispersion

Central tendency is single value that describes the set of data by identifying the central position of the data set.

In statistics, this is the typical value of a probability distribution.

$$A = \frac{1}{n}\sum_{i=1}^{n} a_i = \frac{a_1 + a_2 + a_3 + ... + a_n}{n}$$

Arithmetic mean

$$A = \frac{\sum_{i=1}^{n} w_i f_i}{\sum_{i=1}^{n} w_i}$$

Weighted arithmetic mean

Central tendencies and dispersion

Median: It is the positional value that divides the distribution in two equal parts.

1, 2, 3, 4, 5, 6, 7, 8, 9

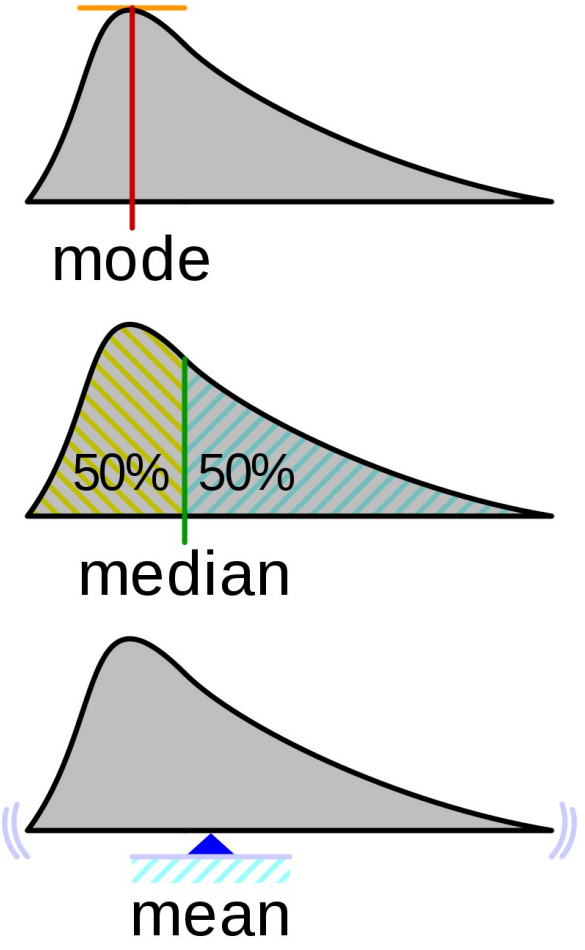
Median

1, 2, 3, 4, 5, 6, 7, 8, 9, 10

Median = (5+6)/2

Mode: It is typical value around which distribution is concentrated or most frequent value in the data set.

1, 3, 6, 6, 6, 6, 7, 7, 12, 12, 17



Central tendencies and dispersion

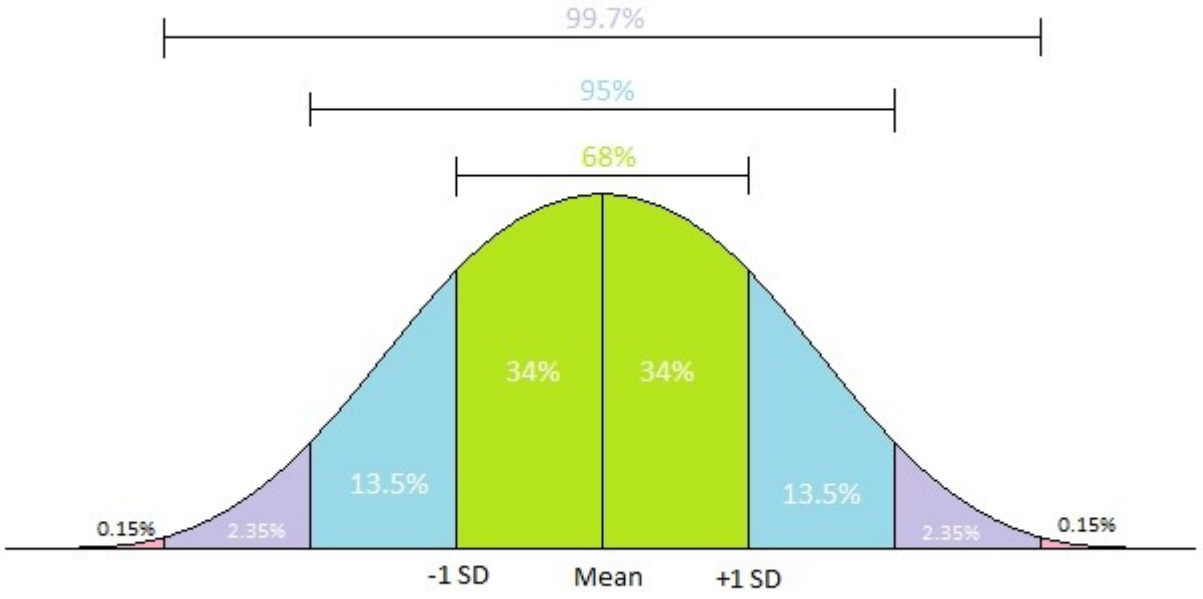
Variance: It is a measurement of the spread between numbers in a data set.

$$S^2 = \frac{\sum_{i=1}^{n} (x_i - \bar{x})^2}{n-1}$$

Standard deviation: It measures the dispersion of a data set relative to its mean.

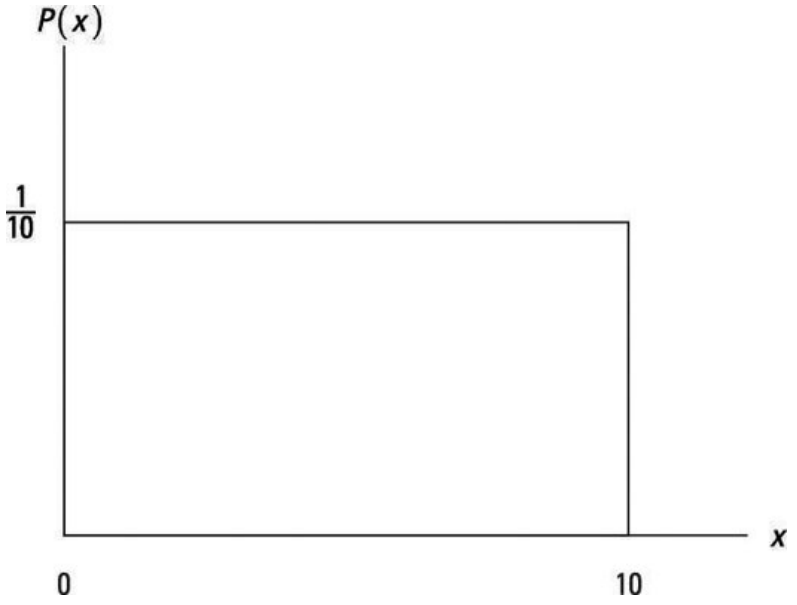
$$\sigma = \sqrt{S^2} = \sqrt{\frac{\sum_{i=1}^{n} (x_i - \bar{x})^2}{n-1}}$$

Central tendencies and dispersion



Frequency distribution

Probabiltiy density function : It is a function that provides the relative likelyhood of the random variable at that given point.



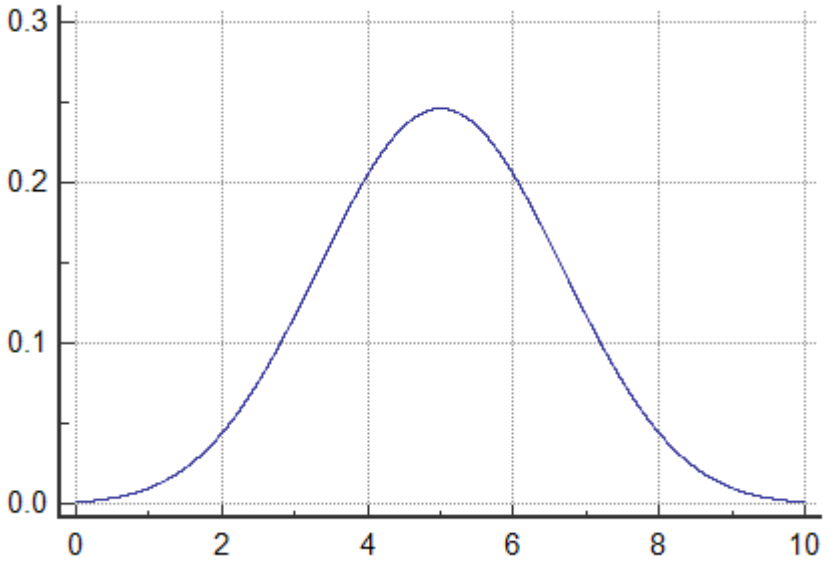
Continious unifrom distribution

$$P(x) = \begin{matrix} \frac{1}{b-a} & for\ a \le x \le b \\ 0 & elsewhere \end{matrix}$$

$$\bar{x} = \frac{a+b}{2}$$

$$\sigma^2 = \frac{b-a}{12}$$

Frequency distribution



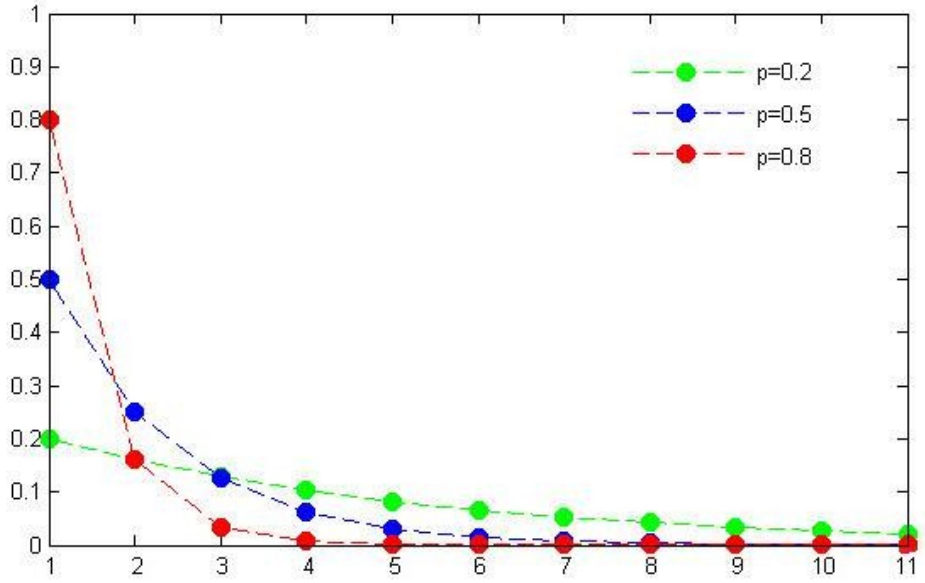
Binomial distribution

$$P(x)=\frac{n!}{(n-x)!\,x!}\,p^{x}q^{n-x}$$

$$\bar{x}=np$$

$$\sigma^{2}=np(1-p)$$

Frequency distribution



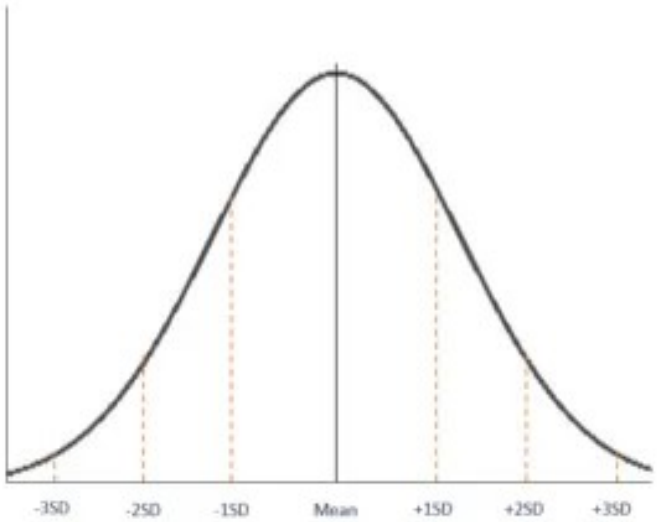
Geometric distribution

$$P(x)=(1-p)^{x-1}p$$

$$\bar{x}=\frac{1}{p}$$

$$\sigma^2=\frac{1-p}{p^2}$$

Frequency distribution



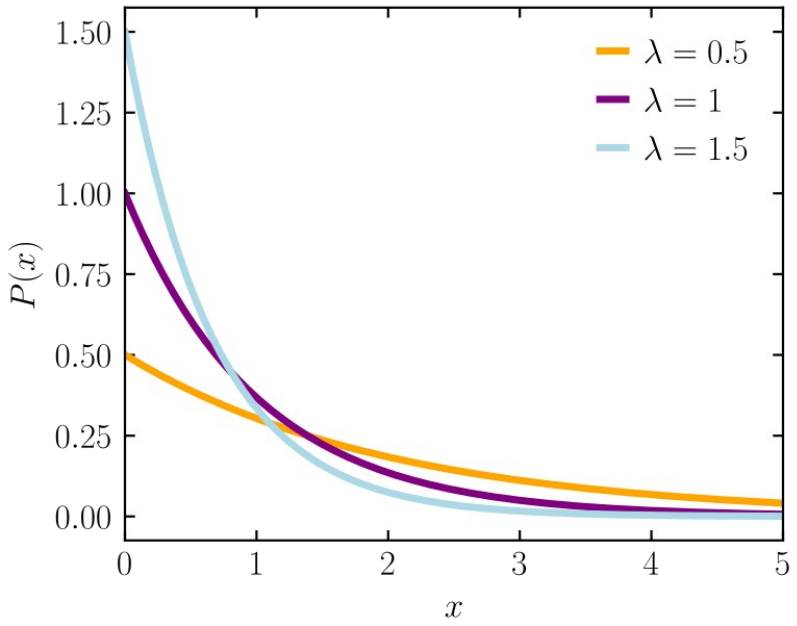
Normal distribution

$$P(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{\frac{-1}{2\pi}\left(\frac{x-\mu}{\sigma}\right)^2}$$

$$\bar{x} = \mu$$

$$\sigma^2 = \sigma^2$$

Frequency distribution



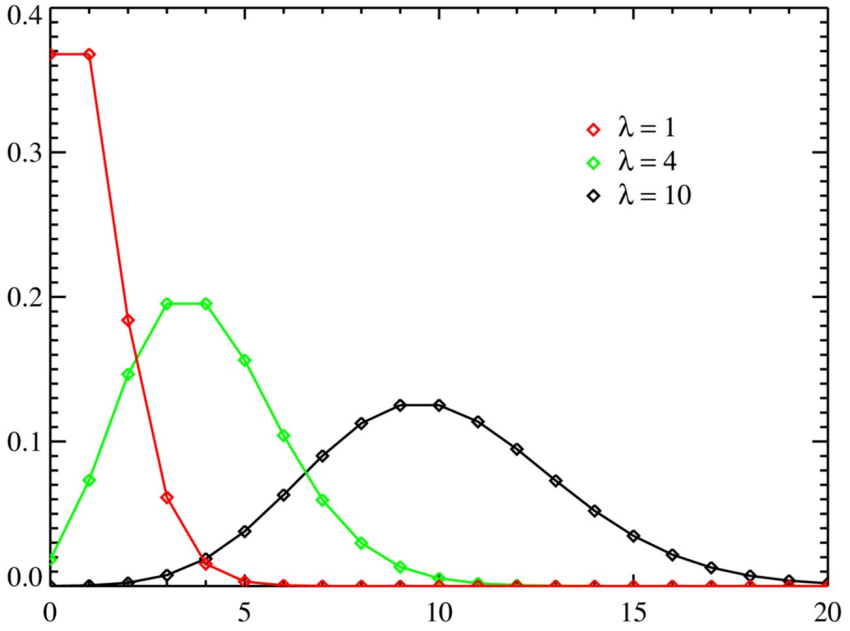
Exponential distribution

$$P(x) = \lambda e^{-\lambda x}$$

$$\bar{x} = \frac{1}{\lambda}$$

$$\sigma^2 = \frac{1}{\lambda^2}$$

Frequency distribution



Poisson distribution

$$P(x) = \frac{\lambda^x e^{-\lambda}}{x!}$$

$$\bar{x} = \lambda$$

$$\sigma^2 = \lambda$$

Error analysis

Errors are inevitable in computer simulations.

When simulating a phenomena, error can occur due to humans or computer

Approximation errors: This arises due to the simplification of mathematical functions, in order to solve them in computers. This includes replacement of infinite series by finite sums, infinitesimal intervals by finite ones, and variable functions by constants.

$$\sin(x) = \frac{\sum_{n=0}^{\infty} (-1)^n x^{2n+1}}{(2n+1)!} \approx \frac{\sum_{n=0}^{N} (-1)^n x^{2n+1}}{(2n+1)!}$$

$$\longrightarrow \quad \sin(x) + \zeta(x, N)$$

For every reasonable approximation, the approximation error should decrease as N increases.

Error analysis

Round-off errors: This is the imprecision arising from the finite number of digits used to store floating-point numbers.

The overall round-off error accumulates as the computer handles more numbers, that is, as the number of steps in a computation increases, and may cause some algorithms to become unstable with a rapid increase in error.

$$\frac{1}{3} = 0.33333333$$

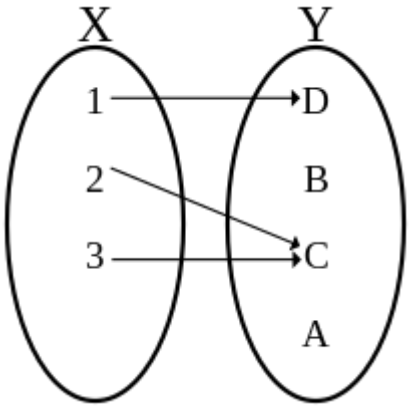
Computer representation x_c of the exact number x is

$$x_c = x\left(1 + \epsilon_x\right)$$
$$\epsilon_x = \epsilon_{\text{approx}} + \epsilon_{roundoff}$$

Simple functions

Function: It is a relationship between two sets.

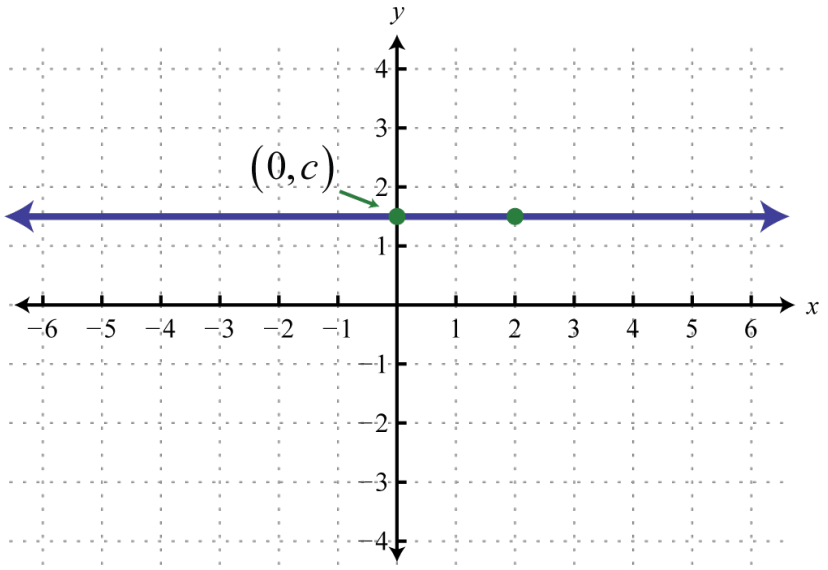
y ---> {x}



Domain Range

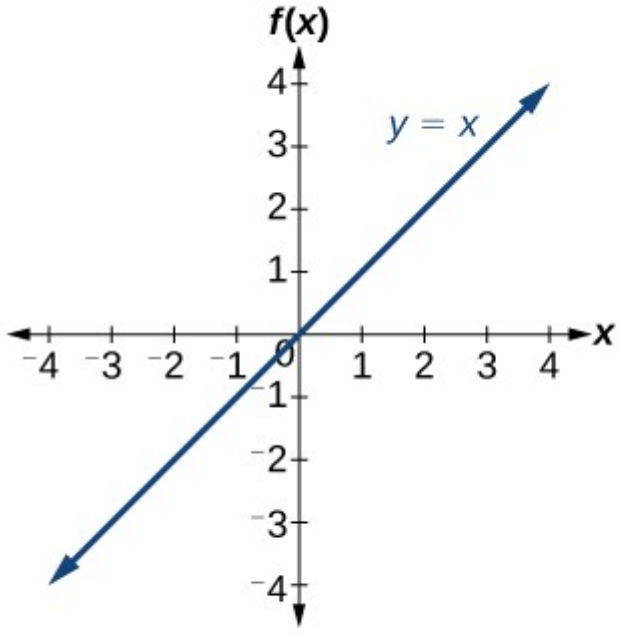
$f(x) = c$

x	$f(x)$
0	c
2	c



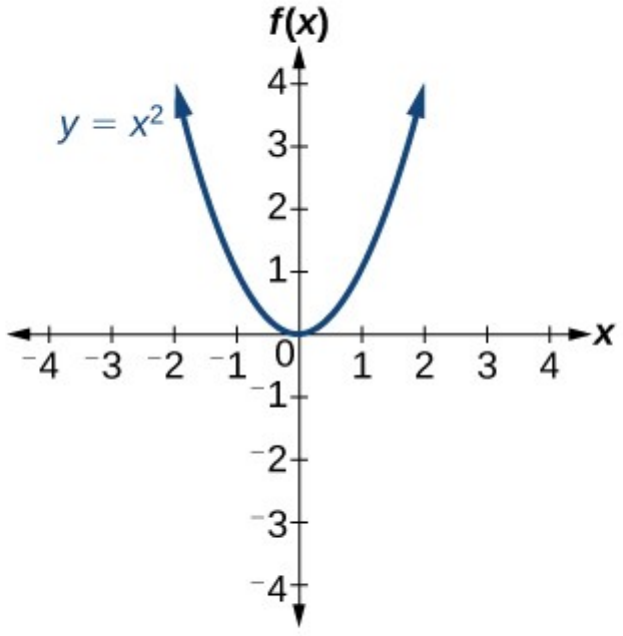
Simple functions

Identity



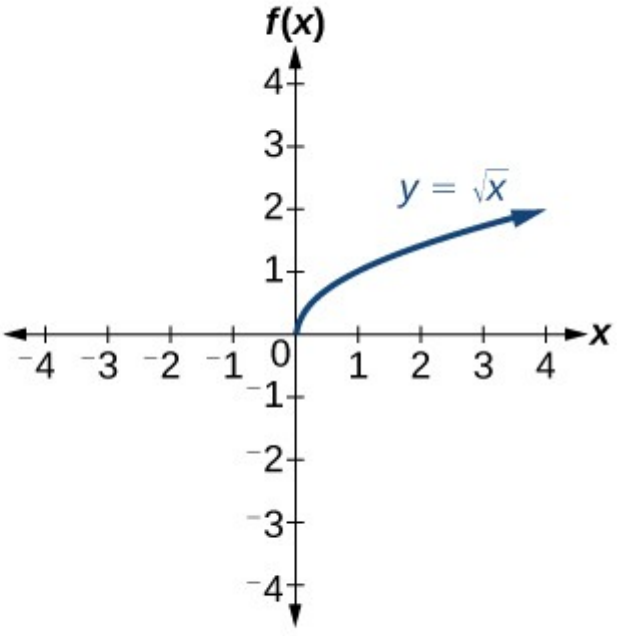
Domain: $(-\infty, \infty)$
Range: $(-\infty, \infty)$

Square



Domain: $(-\infty, \infty)$
Range: $[0, \infty)$

Square Root



Domain: $[0, \infty)$
Range: $[0, \infty)$

Simple functions

Find the domain and range of the functions

$$y=\frac{1}{x-3}-5$$

$$y=\frac{x^2-3x-4}{x+1}$$

Derivative

Derivative of a function f(x) is defined as

$$\frac{dy}{dx} = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

$$m = \lim_{x \to a} \frac{f(x) - f(a)}{x \to a}$$

Find the derivative of

$$f(x) = 2x^2 + 3x - 5 \; at \; x = -1$$

$$f(x) = \sin(x)$$

Derivative

Prove the following

1. Power rule
2. Product rule
3. Quotient rule

Basic Derivatives Rules

Constant Rule: $\frac{d}{dx}(c) = 0$

Constant Multiple Rule: $\frac{d}{dx}[cf(x)] = cf'(x)$

Power Rule: $\frac{d}{dx}(x^n) = nx^{n-1}$

Sum Rule: $\frac{d}{dx}[f(x) + g(x)] = f'(x) + g'(x)$

Difference Rule: $\frac{d}{dx}[f(x) - g(x)] = f'(x) - g'(x)$

Product Rule: $\frac{d}{dx}[f(x)g(x)] = f(x)g'(x) + g(x)f'(x)$

Quotient Rule: $\frac{d}{dx}\left[\frac{f(x)}{g(x)}\right] = \frac{g(x)f'(x) - f(x)g'(x)}{[g(x)]^2}$

Chain Rule: $\frac{d}{dx}f(g(x)) = f'(g(x))g'(x)$

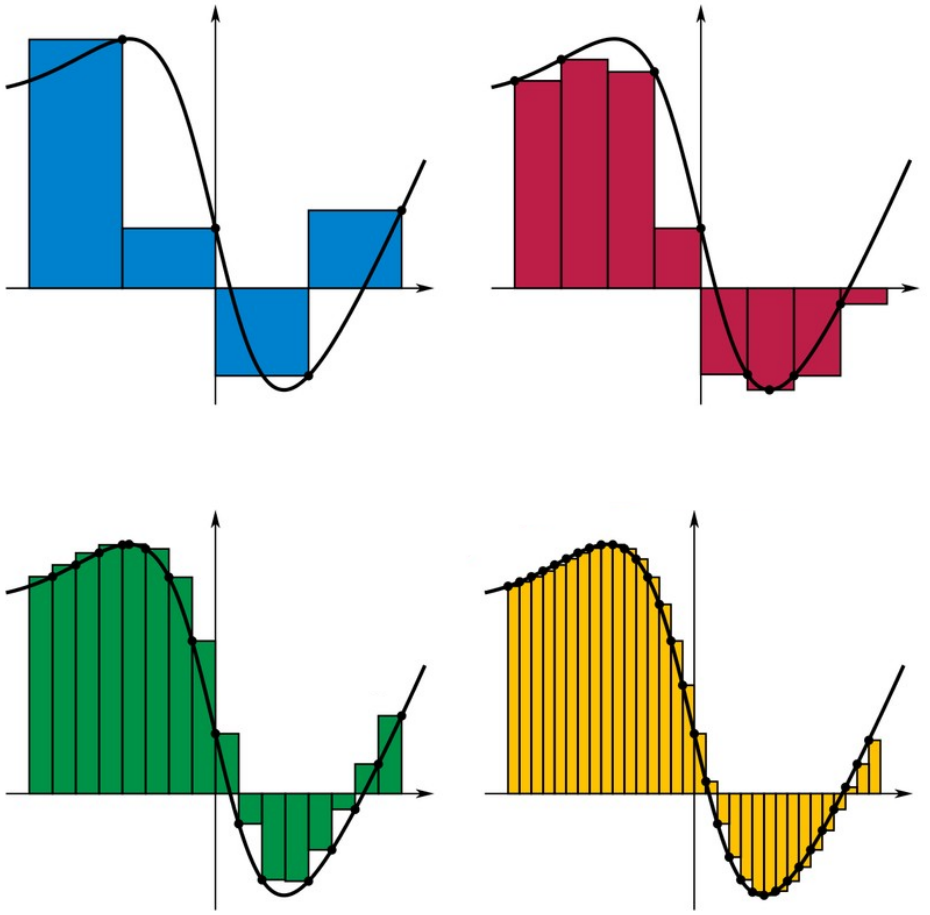
Integral

It is the inverse process of differentiation.

$$f(x)=\frac{dF(x)}{dx}$$

$$\int f(x)\,dx=F(x)+C$$

$$\int f(x)\,dx=\sum_{i=0}^{n-1} f(t_1)(x_{i+1}-x_i)$$



Integral

Integrate the following

$$\int \frac{dx}{1+\tan(x)}$$

$$\int \sin^3(x)\cos^2(x)\,dx$$

Numerical differentiation

$$f'(x)=\lim_{h\to 0}\frac{f(x+h)-f(x)}{h}$$ Forward difference

$$f'(x)=\lim_{h\to 0}\frac{f(x)-f(x-h)}{h}$$ Backward difference

$$f'(x)=\lim_{h\to 0}\frac{f(x+h)-f(x-h)}{2h}$$ Central difference

Numerical differentiation

Calculate the first derivative of cos(x) at x = π/3 for

a) h = 0.1

b) h = 0.01

c) h = 0.001

d) h = 0.0001

Also calculate error in each case.

h	forward	Error forward	backward	Error backward	central	Error central
0.1						
0.01						
0.001						
0.0001						

(cos(x+h) -cos(x))/h, x=pi/3
a) 0.1

f'_first_principle (x=pi/3) = (cos(pi/3+0.1) – cos(pi))/0.1

-0.88956192323

f'_direct(pi) = -sin(pi/3) = $3^{0.5}$/2 = -0.86602540378

Error = f'_direct(x=pi/3) - f'_first_principle (x=pi/3)
= -0.86602540378 + 0.88956192323
= -0.02353651945

Numerical integration

$$I = \int_a^b f(x)dx \approx \sum_{i=1}^{n} w_i \ f(x_i)$$

Integrate the following $\int_0^1 \cos(x)\,dx$ for w_i

a) 0.1
b) 0.01
c) 0.001
d) 0.0001

Also calculate error in each case.

$$\int_0^1 \cos(x)\,dx = w\sum_{i=1}^{10} \cos(x_i) = w*(\cos(0)+\cos(0.1)+\cos(0.2)+\cos(0.3)+....+\cos(0.9))$$

$$\int_0^1 \cos(x)\,dx = w\sum_{i=1}^{10} \cos(x_i) = w*(\cos(0.1)+\cos(0.2)+\cos(0.3)+....+\cos(1.0))$$

$$\int_0^1 \cos(x)\,dx = w\sum_{i=1}^{10} \cos(x_i) = w*(\cos(0)+\cos(0.1))/2+(\cos(0.1)+\cos(0.2))/2+..$$

$$I = \int_a^b f(x)dx \approx \sum_{i=1}^{n} w_i \ f(x_i)$$

$$\int_a^b f(x)\,dx = w * \sum_{i=1}^{n} f(x_i) = w*((f(a)+f(a+w))/2+(f(a+w)+f(a+2w))/2+(f(a+2w)+f(a+3w))/2+\ldots)$$

$$= w*((f(a)+f(b))/2+\sum_{i=1}^{n-1} f(x_i))$$

Functions - Python

```
import matplotlib.pyplot as plt
import numpy as np
x = np.linspace(-5,5,100)
y = 2*x+1
plt.plot(x, y, label='y=2x+1')
plt.title('Graph of y=2x+1')
plt.xlabel('x')
plt.ylabel('y')
plt.legend(loc='upper left')
plt.grid()
plt.show()
```

Functions - Python

```
import matplotlib.pyplot as plt
import numpy as np
fig = plt.figure()
x = np.linspace(-5,5,100)
plt.plot(x, 2*x+1, '-r', label='y=2x+1')
plt.plot(x, 2*x-1,'-.g', label='y=2x-1')
plt.plot(x, 2*x+3,':b', label='y=2x+3')
plt.plot(x, 2*x-3,'--m', label='y=2x-3')
plt.legend(loc='upper left')
plt.show()
```

Functions - Python

```
import matplotlib.pyplot as plt
import numpy as np
fig = plt.figure()
ax = fig.add_subplot(1, 1, 1)
x = np.linspace(-5,5,100)
ax.spines['left'].set_position('center')
ax.spines['bottom'].set_position('center')
ax.spines['right'].set_color('none')
ax.spines['top'].set_color('none')
ax.xaxis.set_ticks_position('bottom')
ax.yaxis.set_ticks_position('left')
plt.plot(x, 2*x+1, 'r', label='y=2x+1')
plt.plot(x, 2*x-1,'g', label='y=2x-1')
plt.plot(x, 2*x+3,'b', label='y=2x+3')
plt.plot(x, 2*x-3,'k', label='y=2x-3')
plt.legend(loc='upper left')
plt.show()
```

Functions - Python

```
import matplotlib.pyplot as plt
import numpy as np

x = np.linspace(-2, 2, 100)
y = x ** 2

fig = plt.figure(figsize = (10, 5))
plt.plot(x, y)
plt.show()
```

Functions - Python

```
import matplotlib.pyplot as plt
import numpy as np

x = np.linspace(-6, 6, 50)
y = np.cos(x)
plt.plot(x, y, 'b', label ='cos(x)')
plt.xlabel('x-axis')
plt.ylabel('y-axis')
```

Functions - Python

```
import matplotlib.pyplot as plt
import numpy as np

x = np.linspace(0, 1, 1000)
y = np.exp(x)
plt.plot(x, y, 'b', label ='exp(x)')
plt.xlabel('x-axis')
plt.ylabel('y-axis')
```

Functions - Python

```
import matplotlib.pyplot as plt
import numpy as np

x = np.linspace(1, 100, 1000)
y = np.log(x)
plt.plot(x, y, 'b', label ='ln(x)')
plt.xlabel('x-axis')
plt.ylabel('y-axis')
```

Functions - Python

```
import matplotlib.pyplot as plt
import numpy as np

x = np.linspace(1, 100, 1000)
y = np.log10(x)
plt.plot(x, y, 'b', label ='log(x)')
plt.xlabel('x-axis')
plt.ylabel('y-axis')
```

Central tendencies - Python

```
import numpy as np
import statistics

data1 = [1, 3, 4, 5, 7, 9, 2]

x = statistics.mean(data1)

print("Mean is :", x)
```

Central tendencies - Python

```
import numpy as np
import statistics
from statistics import mean

data1 = [1, 3, 4, 5, 7, 9, 2]

print("Mean is :", mean(data1))
```

Central tendencies - Python

```
import numpy as np
import statistics

data1 = [1, 3, 4, 5, 7, 9, 2]

print("Median of data-set is : ",  statistics.median(data1)))
```

Central tendencies - Python

```
import statistics

set1 =[1, 2, 3, 3, 4, 4, 4, 5, 5, 6]

print("Mode of given data set is :", (statistics.mode(set1)))
```

Central tendencies - Python

```
import statistics

sample = [1, 2, 3, 4, 5]

print("Standard Deviation of sample is:" , (statistics.stdev(sample)))
```

Random numbers - Python

```
from numpy.random import rand
for i in range(10):
 print(rand())
```

```
from numpy.random import randint
for i in range(10):
 print(randint(0,10))
```

Random numbers - Python

```
from numpy.random import rand

l=[]
for i in range(10):
 l.append(rand())


print(l)

plt.plot(l)
plt.show()
```